

Limit Order Book Research

From Microstructure to Executable Signals

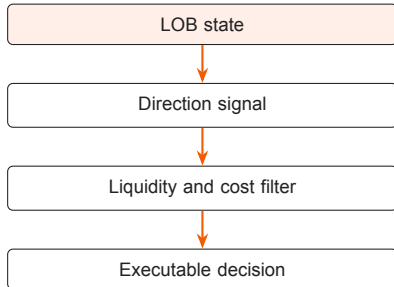
15-minute research briefing | June 4, 2026

■ Executive Message

Core claim

LOB research should be evaluated as an **execution problem**, not only a prediction problem.

- ▶ Price direction is useful only if it survives spread, impact, latency, and inventory risk.
- ▶ The order book gives a tradable market state: queue depth, imbalance, liquidity, and resiliency.
- ▶ Our internal study should start with liquidity-aware diagnostics before complex models.



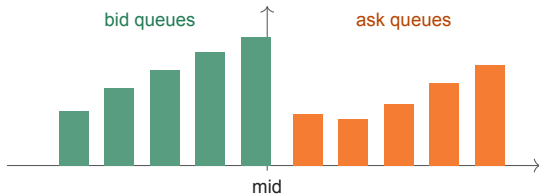
■ What the Book Adds Beyond Price

- ▶ A LOB records resting buy and sell queues across price levels.
- ▶ A typical model input is a recent history of the first L levels:

$$X_t \in \mathbb{R}^{T \times 4L}.$$

- ▶ Queue depletion explains why the next price change happens.
- ▶ Queue imbalance gives a first-order pressure measure:

$$I_t^{(k)} = \frac{\sum_{i=1}^k q_i^b - \sum_{i=1}^k q_i^a}{\sum_{i=1}^k q_i^b + \sum_{i=1}^k q_i^a}.$$



Practical meaning

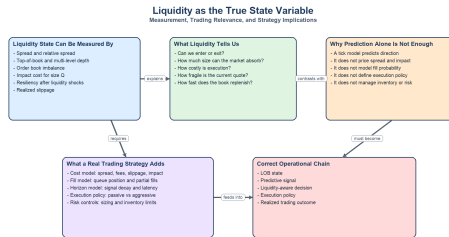
The market state is a short movie of queue shapes, not a price line.

■ Liquidity Is the Tradable State

A trade is viable only when

$$E[\Delta p] > \text{spread} + \text{fee} + \text{impact} + \text{delay risk}.$$

- ▶ Spread measures immediate trading friction.
- ▶ Depth and impact determine capacity.
- ▶ Resiliency determines how quickly the book recovers.
- ▶ Slippage is the operational truth of the signal.

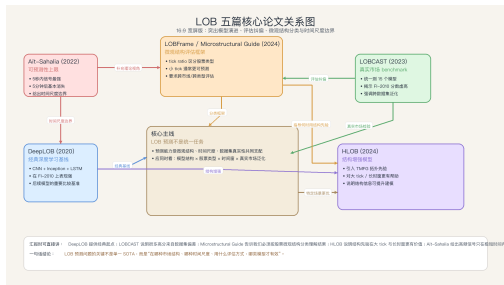


Bottom line: sustainable P&L requires both prediction and a measurable liquidity/protection model.

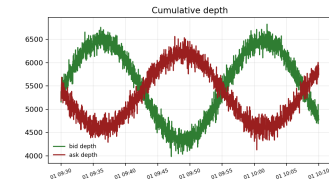
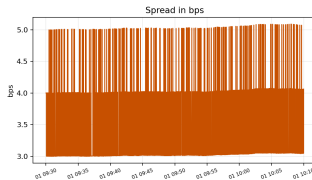
■ What the Literature Suggests

- ▶ Queueing and Hawkes models explain microstructure mechanics.
- ▶ DeepLOB shows that book snapshots can carry short-horizon signal.
- ▶ Recent benchmarks warn that evaluation can be fragile across datasets, horizons, and market regimes.
- ▶ For our use case, the first milestone is not the most complex model; it is a reliable, cost-aware research pipeline.

Source: Cont and de Larrard; Bacry et al.; Zhang et al.; recent LOB benchmark studies.



■ Internal Data Diagnostics

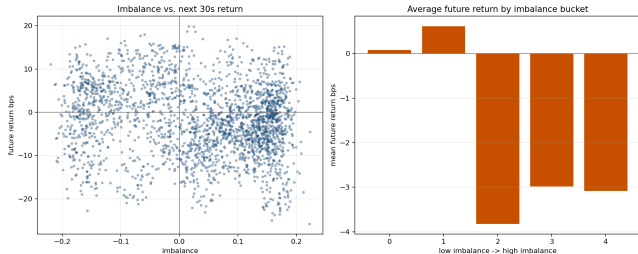


Current snapshot

Rows	2,400
Median spread	3.05 bps
Median depth	10,865
Impact proxy	1.62 bps

- ▶ Replace placeholder with intranet-generated figures.
- ▶ Focus on spread, depth, imbalance, and impact stability.

■ From Diagnostics to Signal



Interpretation

- ▶ If imbalance has no stable relation with future return, it is not a standalone signal.
- ▶ If it works only under tight-spread regimes, it should be conditional.
- ▶ The final score is execution-aware P&L, not classification accuracy.

■ Proposed Next Step

Two-week plan

- ▶ Week 1: data quality, feature library, baseline diagnostics.
- ▶ Week 2: cost-aware backtest, liquidity filters, first strategy prototype.
- ▶ DeepLOB / Transformer benchmark after the simple baseline is stable.

Decision needed

- ▶ Priority instruments and venues.
- ▶ First execution assumption: aggressive, passive, or hybrid.
- ▶ Minimum capacity threshold for commercial relevance.

Build the execution-aware pipeline first; add deep models after the baseline is trustworthy.